

# HBTU KANPUR — B.TECH ENGINEERING CHEMISTRY

## Unit 4 Complete Study Guide: Stereochemistry, Reaction Mechanisms, Named Rearrangements & Environmental Effluent Analysis

**Core Curriculum:** Stereoisomerism (1 & 2 chiral centers, E/Z, R/S, Atropisomerism, Allenes, Spiranes, CD), Electronic Effects, Intermediates, SN/SE mechanisms, 5 Named Reactions (Suzuki, Fries, Wagner-Meerwein, Umpolung, Favorskii)

**Environmental Section:** Effluent Gas Analysis (SO<sub>x</sub>, NO<sub>x</sub>, H<sub>2</sub>S), Liquid Effluent (DO, BOD, COD), Pollution Control & Ozone Layer Depletion

### PART 1 STEREOCHEMISTRY & MOLECULAR CHIRALITY

#### 1. Stereoisomerism in Compounds with One & Two Chiral Centers

**Stereoisomerism** occurs when molecules share identical molecular formulas and constitution (connectivity of atoms) but differ in the 3D spatial arrangement of their atoms in space. A carbon atom bonded to four chemically distinct ligands is designated an **asymmetric or stereogenic center (chiral center)**.

##### Compounds with One Chiral Center

- Maximum number of stereoisomers =  $2^n = 2^1 = 2$ .
- Generates a pair of non-superimposable mirror images termed **enantiomers** (e.g., D- and L-Lactic acid:  $\text{CH}_3\text{-C}^*\text{H(OH)-COOH}$ ).
- An equimolar (1:1) mixture of enantiomers is an optically inactive **racemic mixture ( $\pm$ )** due to external compensation.

##### Compounds with Two Chiral Centers

- Dissimilar Ends ( $\text{R-C}^*\text{Hab-C}^*\text{Hcd-R}'$ ):** Number of stereoisomers =  $2^2 = 4$  (2 pairs of enantiomers; diastereomeric relationship between pairs). Example: 2-bromo-3-chlorobutane.
- Similar Ends ( $\text{R-C}^*\text{Hab-C}^*\text{Hab-R}$ ):** Example: Tartaric acid ( $\text{HOOC-CH(OH)-CH(OH)-COOH}$ ). Total isomers = 3: pair of enantiomers ((+)-tartaric acid, (–)-tartaric acid) + one **meso-tartaric acid** (optically inactive due to internal compensation via an internal plane of symmetry,  $\sigma$ ).

#### 2. Enantiomers vs. Diastereomers

| Property                   | Enantiomers                                                                                                              | Diastereomers                                                                                                   |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Definition</b>          | Stereoisomers that are non-superimposable mirror images of one another.                                                  | Stereoisomers that are <b>not</b> mirror images of one another.                                                 |
| <b>Spatial Relation</b>    | Inversion of configuration at <i>all</i> chiral centers ( $2\text{R}, 3\text{R} \leftrightarrow 2\text{S}, 3\text{S}$ ). | Inversion at some, but not all, chiral centers ( $2\text{R}, 3\text{R} \leftrightarrow 2\text{R}, 3\text{S}$ ). |
| <b>Physical Properties</b> | Identical melting points, boiling points, densities, refractive indices, and dipole moments in achiral media.            | Distinct physical properties (different m.p., b.p., solubilities, dielectric constants, and dipole moments).    |
| <b>Optical Rotation</b>    | Equal in magnitude but opposite in sign (+ / d and – / l).                                                               | Different magnitudes and signs; one can even be an optically inactive meso form.                                |

| Property                       | Enantiomers                                                                                                | Diastereomers                                                                                                               |
|--------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Separation (Resolution)</b> | Cannot be separated by fractional distillation; requires chiral resolving agents or chiral chromatography. | Easily separated by conventional physical techniques (fractional crystallization, fractional distillation, chromatography). |

### 3. E-Z Nomenclature & R-S Configuration

#### A. E-Z Nomenclature for Geometrical Isomers

The classical cis/trans notation fails for tri- and tetra-substituted alkenes. The IUPAC **E-Z system** utilizes the **Cahn-Ingold-Prelog (CIP) priority rules** to rank the two groups attached to each  $sp^2$  carbon:

- **(Z)-Isomer (Zusammen = together):** Higher priority groups on each double-bond carbon are situated on the *same side* of the double bond axis.
- **(E)-Isomer (Entgegen = opposite):** Higher priority groups are situated on *opposite sides* of the double bond axis.
- **Example:** 1-bromo-1-chloro-2-fluoroethene (**BrClC=CHF**). At C1: **Br** (Z=35) > **Cl** (Z=17). At C2: **F** (Z=9) > **H** (Z=1). If **Br** and **F** are on the same side, it is **(Z)**; if opposite, **(E)**.

#### B. R-S Absolute Configuration (CIP Sequence Rules)

1. **Priority Assignment:** Atoms directly attached to the chiral center are assigned priority 1 (highest) to 4 (lowest) based on decreasing atomic number (Z):  $-I > -Br > -Cl > -SO_3H > -F > -OH > -NH_2 > -COOH > -CHO > -CH_2OH > -CH_3 > -H$ . Isotopes: Higher mass takes priority ( $^3T > ^2D > ^1H$ ).
2. **Tie-Breaking:** If directly attached atoms are identical, compare atoms at the second, third, etc., points of difference along the chain.
3. **Multiple Bonds:** Doubly or triply bonded atoms are treated as equivalent to duplicate/triplicate bonded phantom atoms (e.g.,  $-CH=O$  is treated as  $-CH(O)_2$ ;  $-C\equiv N$  as  $-C(N)_3$ ).
4. **Orientation Rule (3D / Wedge-Dash):** View the chiral center with the lowest priority group (4) pointing directly away from the observer (on a dash). If priority  $1 \rightarrow 2 \rightarrow 3$  proceeds **clockwise**, the configuration is **R** (Rectus). If **counterclockwise**, it is **S** (Sinister).
5. **Fischer Projection Rule:** If group 4 is on a *vertical bond*, clockwise = **R**, counterclockwise = **S**. If group 4 is on a *horizontal bond*, reverse the outcome: clockwise = **S**, counterclockwise = **R**.

### 4. Atropisomerism & Optical Activity in Molecules Lacking Chiral Centers

Molecules can exhibit chirality in the absence of an asymmetric  $sp^3$  stereocenter if they possess a **chiral axis** or **chiral plane** that lacks an alternating axis of symmetry ( $S_n$ ), an inversion center (i), and a plane of symmetry ( $\sigma$ ).

#### A. Atropisomerism in Biphenyls

**Atropisomerism** is conformational stereoisomerism resulting from *restricted rotation around a single bond* due to severe steric hindrance of bulky substituents at the ortho-positions.

##### Conditions for Chiral Atropisomers in Biphenyls:

- Rotation about the central C-C pivot single bond must be kinetically hindered at room temperature (activation barrier to rotation  $\Delta G^\ddagger > 100 \text{ kJ/mol}$ , half-life  $> 1000 \text{ s}$ ). Bulky ortho groups (e.g.,  $-NO_2$ ,  $-COOH$ ,  $-SO_3H$ ,  $-I$ ,  $-Br$ ) force the two aromatic rings into mutually perpendicular orthogonal planes.
- Each individual phenyl ring must be *unsymmetrically substituted* in the ortho/meta positions ( $a \neq b$  and  $c \neq d$ ).  
Example: **6,6'-dinitrobiphenyl-2,2'-dicarboxylic acid**. The molecule has no plane or center of inversion; it exists as separable, stable enantiomeric atropisomers.

## B. Optical Isomerism in Allenes

Allenenes contain cumulated double bonds:  $C=C=C$ . The central carbon is  $sp$  hybridized with two orthogonal p orbitals ( $p_y$ ,  $p_z$ ) forming mutually perpendicular  $\pi$ -bonds with terminal  $sp^2$  carbons.

- As a consequence, the two sets of terminal substituents lie in mutually perpendicular planes.
- **Chirality Condition:** An allene of type  $abC=C=Cab$  or  $abC=C=Ccd$  is chiral and optically active if neither terminal carbon bears two identical groups ( $a \neq b$ ). It possesses an axis of chirality ( $C_2$  axis). Example: Penta-2,3-diene ( $CH_3CH=C=CHCH_3$ ).

## C. Optical Isomerism in Spiranes

Spiranes consist of two carbocyclic rings joined at a single shared  $sp^3$  quaternary spiro-carbon atom.

- Due to tetrahedral geometry at the shared junction, the planes of the two rings are orthogonal (at  $90^\circ$  to one another), exactly analogous to allenes.
- If the terminal ends of the two perpendicular rings are unsymmetrically substituted ( $a \neq b$  on both rings), the plane of symmetry is eliminated, producing optically active enantiomers (e.g., 2,6-diaminospiro[3.3]heptane).

## 5. Circular Dichroism (CD) & Optical Rotatory Dispersion (ORD)

When plane-polarized light passes through an optically active chiral medium containing a chromophore, it interacts with two circularly polarized components: right-circularly polarized (RCP) and left-circularly polarized (LCP) light.

### Circular Dichroism (CD)

- **Principle:** Differential absorption of LCP and RCP light by a chiral chromophore:  $\Delta A = A_L - A_R$ .
- **Molar Circular Dichroism:**  $\Delta \epsilon = \epsilon_L - \epsilon_R$ .
- The emergent light becomes elliptically polarized. Ellipticity:  $\theta = 32.98 \cdot \Delta \epsilon$  (degrees).
- **Cotton Effect:** The anomalous dispersion curve in the region of an absorption band. If  $\epsilon_L > \epsilon_R$  at peak, it gives a **positive Cotton effect**; if  $\epsilon_R > \epsilon_L$ , a **negative Cotton effect**.

### Applications of CD / ORD

- **Absolute Configuration:** Determining absolute stereochemistry using empirical sector rules (e.g., Octant Rule for chiral cyclohexanones).
- **Biochemical Conformation:** Dissecting protein secondary structure (distinct far-UV CD signatures for  $\alpha$ -helices with negative minima at 222 & 208 nm,  $\beta$ -sheets at 217 nm, and random coils).
- **Monitoring:** Tracking conformational transitions, drug-receptor binding, and denaturation.

## PART 2 ELECTRONIC EFFECTS, INTERMEDIATES & REACTION MECHANISMS

### 1. Electronic Effects in Organic Molecules

- **Inductive Effect (I-Effect):** Permanent displacement of  $\sigma$ -electrons along a carbon chain caused by electronegativity differences. Transmitted through  $\sigma$ -bonds and attenuates rapidly, becoming negligible beyond 3 bonds.
  - *-I Effect (Electron-withdrawing):*  $-NR_3^+ > -NO_2 > -SO_3H > -CN > -COOH > -F > -Cl > -Br > -I > -OH > -C_6H_5 > -H$ . Increases carboxylic acid strength (e.g.,  $CCl_3COOH > CHCl_2COOH > CH_2ClCOOH > CH_3COOH$ ).
  - *+I Effect (Electron-releasing):*  $-O^- > -COO^- > -C(CH_3)_3 > -CH(CH_3)_2 > -CH_2CH_3 > -CH_3 > -D > -H$ . Stabilizes carbocations; destabilizes carbanions.
- **Electromeric Effect (E-Effect):** A temporary, dynamic electronic displacement occurring exclusively at the demand of an attacking reagent in multiple bonds ( $C=C$ ,  $C=O$ ,  $C \equiv N$ ). Disappears upon reagent removal.
  - *+E Effect:*  $\pi$ -electron pair shifts toward the atom to which the attacking electrophile attaches (e.g., addition of  $H^+$  to alkene).

- **-E Effect:**  $\pi$ -electron pair shifts away from the atom to which the attacking nucleophile attaches (e.g., addition of  $\text{CN}^-$  to carbonyl carbon:  $>\text{C}=\text{O} + \text{CN}^- \rightarrow >\text{C}(\text{CN})-\text{O}^-$ ).
- **Mesomeric / Resonance Effect (M-Effect):** Permanent delocalization of  $\pi$ -electrons or lone pairs through conjugated  $\pi$ -systems. Independent of distance along the conjugated backbone.
  - **+M Effect (Electron-donating by resonance):** Groups with unshared lone pairs conjugated with a  $\pi$ -system:  $-\text{O}^- > -\text{NR}_2 > -\text{NHR} > -\text{NH}_2 > -\text{OH} > -\text{OR} > -\text{NHCOR} > -\text{halogen}$ . Activates benzene ring towards electrophiles and directs ortho/para.
  - **-M Effect (Electron-withdrawing by resonance):** Groups with multiple bonds to electronegative atoms conjugated to a  $\pi$ -system:  $-\text{NO}_2 > -\text{CN} > -\text{SO}_3\text{H} > -\text{CHO} > -\text{COR} > -\text{COOH} > -\text{COOR}$ . Deactivates benzene ring and directs meta.

## 2. Study of Reaction Intermediates

| Intermediate                                     | Structure & Hybridization                                                                                                                                                                                                                           | Stability Order & Factors                                                                                                                                                                                                                                                             | Generation Methods                                                                                                                                                                                                                                                                                           | Characteristic Reactions                                                                                                                                                                                                                                       |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Carbocation</b><br>( $\text{R}_3\text{C}^+$ ) | Planar trigonal, $\text{sp}^2$ , 6 valence electrons, vacant unhybridized p-orbital.                                                                                                                                                                | $3^\circ > 2^\circ > 1^\circ > \text{CH}_3^+$ .<br>Stabilized by hyperconjugation ( $\alpha$ -C-H bonds), +I effect, allylic/benzylic resonance, tropylium aromaticity.                                                                                                               | <ul style="list-style-type: none"> <li>• Heterolytic cleavage of <math>\text{R-X}</math>.</li> <li>• Protonation of alkenes (<math>\text{R-CH=CH}_2 + \text{H}^+</math>).</li> <li>• Diazotization of primary amines (<math>\text{R-NH}_2 + \text{HNO}_2</math>).</li> </ul>                                 | <ul style="list-style-type: none"> <li>• Nucleophilic capture (<math>\text{S}_{\text{N}}1</math>).</li> <li>• Proton loss forming alkene (<math>\text{E1}</math>).</li> <li>• 1,2-hydride/alkyl <b>Wagner-Meerwein rearrangements</b>.</li> </ul>              |
| <b>Carbanion</b><br>( $\text{R}_3\text{C}^-$ )   | Trigonal pyramidal, $\text{sp}^3$ (rapid umbrella inversion), 8 valence electrons with lone pair; $\text{sp}^2$ if conjugated.                                                                                                                      | $\text{CH}_3^- > 1^\circ > 2^\circ > 3^\circ$ .<br>Stabilized by -I, -M groups ( $-\text{NO}_2$ , $-\text{C}=\text{O}$ , $-\text{CN}$ ), and higher s-character: $\text{sp}$ (acetylide) $> \text{sp}^2 > \text{sp}^3$ .                                                              | <ul style="list-style-type: none"> <li>• Deprotonation of acidic C-H by strong base (LDA, <math>\text{NaNH}_2</math>).</li> <li>• Decarboxylation of <math>\beta</math>-keto acids.</li> <li>• Organometallic transmetalation (<math>\text{R-Li}</math>, <math>\text{RMgX}</math>).</li> </ul>               | <ul style="list-style-type: none"> <li>• Nucleophilic attack on carbonyls (Aldol, Claisen, Grignard additions).</li> <li>• <math>\text{S}_{\text{N}}2</math> alkylation.</li> <li>• Michael conjugate addition.</li> </ul>                                     |
| <b>Carbene</b><br>( $:\text{CR}_2$ )             | Neutral divalent carbon, 6 valence electrons.<br>• <i>Singlet</i> : $\text{sp}^2$ , bent, paired electrons in $\text{sp}^2$ , empty p.<br>• <i>Triplet</i> : $\text{sp}$ linear or bent, two unpaired electrons in separate p orbitals (diradical). | Triplet is generally more stable than singlet by Hund's rule (triplet ground state for $:\text{CH}_2$ by $\sim 38$ kJ/mol).<br>Halogenated carbenes ( $:\text{CCl}_2$ ) prefer singlet due to back-bonding lone pair donation: $:\text{CX}_2$ ( $\text{F} > \text{Cl} > \text{Br}$ ). | <ul style="list-style-type: none"> <li>• <math>\alpha</math>-elimination of chloroform: <math>\text{CHCl}_3 + \text{OH}^- \rightarrow :\text{CCl}_2 + \text{H}_2\text{O} + \text{Cl}^-</math>.</li> <li>• Photolysis/thermolysis of diazomethane (<math>\text{CH}_2\text{N}_2</math>) or ketenes.</li> </ul> | <ul style="list-style-type: none"> <li>• Cyclopropanation with alkenes (singlet is stereospecific syn-addition; triplet is non-stereospecific step-wise addition).</li> <li>• Insertion into C-H and O-H bonds.</li> <li>• Reimer-Tiemann reaction.</li> </ul> |

| Intermediate                          | Structure & Hybridization                                                                                                                                                                              | Stability Order & Factors                                                                                            | Generation Methods                                                                                                                                                                  | Characteristic Reactions                                                                                                                                                               |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nitrene</b><br>( $R-N:$ )          | Neutral univalent nitrogen, 6 valence electrons.<br>• Exists in Singlet ( $sp^2$ ) or Triplet ( $sp$ ) states. Triplet is ground state.                                                                | Triplet is more stable. Electron-rich substituents stabilize singlet via mesomeric donation into the vacant orbital. | • Thermolysis or photolysis of alkyl/aryl/acyl azides: $R-N_3 \rightarrow R-N: + N_2$ .<br>• $\alpha$ -elimination of sulfonyl hydroxylamines.                                      | • Aziridination across double bonds.<br>• Intramolecular C-H insertion.<br>• Acyl nitrene rearrangement to isocyanate: <b>Hofmann, Curtius, Lossen, Schmidt</b> .                      |
| <b>Benzyne</b><br>(Arynes, $C_6H_4$ ) | Dehydrobenzene; benzene ring with an extra formal triple bond formed by lateral overlap of two $sp^2$ orbitals in the molecular plane. Highly strained (bond angle $120^\circ$ vs ideal $180^\circ$ ). | Extremely reactive, transient intermediate. Not isolable; stabilized only under matrix isolation at $< 20$ K.        | • Halobenzene + very strong base ( $NaNH_2$ / liq. $NH_3$ ).<br>• Diazotization of anthranilic acid (2-aminobenzoic acid) with isoamyl nitrite.<br>• Fluorobenzene + organolithium. | • Cine-substitution: Nucleophilic attack yielding mixture of ortho/meta substituted anilines.<br>• Powerful dienophile in Diels-Alder trapping: with furan gives 1,4-epoxynaphthalene. |

### 3. Substitution Reaction Mechanisms: Nucleophilic & Electrophilic

#### A. Nucleophilic Substitution Reactions ( $S_N1$ vs. $S_N2$ )

##### $S_N1$ Mechanism (Unimolecular)

- **Kinetics:** Two-step; Rate =  $k[R-X]$  (First order).
- **Step 1 (Rate-determining step):** Slow heterolysis of C-X bond to generate a planar,  $sp^2$  carbocation intermediate.
- **Step 2:** Fast attack of nucleophile from either the front or rear face.
- **Stereochemistry:** Racemization with slight excess of inversion (due to intimate ion-pair shielding).
- **Substrate Reactivity:**  $3^\circ > 2^\circ > 1^\circ > CH_3$ ; favored by polar protic solvents ( $H_2O$ ,  $MeOH$ ) and weak nucleophiles.

##### $S_N2$ Mechanism (Bimolecular)

- **Kinetics:** Concerted single-step; Rate =  $k[R-X][Nu^-]$  (Second order).
- **Transition State:** Backside nucleophilic attack ( $180^\circ$  to leaving group) via a pentacoordinate trigonal bipyramidal transition state ( $[Nu \cdots C \cdots X]^\ddagger$ ).
- **Stereochemistry:** Complete 100% **Walden Inversion** of configuration ( $R \rightarrow S$  or  $S \rightarrow R$ ).
- **Substrate Reactivity:**  $CH_3 > 1^\circ > 2^\circ \gg 3^\circ$  (governed by steric hindrance); favored by polar aprotic solvents (DMSO, DMF, acetone) and strong nucleophiles.

#### B. Electrophilic Aromatic Substitution ( $S_EAr$ ) Mechanism

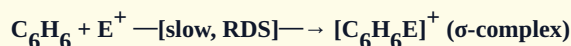
Aromatic systems undergo substitution rather than addition to preserve the high resonance thermodynamic stability of the  $6\pi$  aromatic sextet. The general mechanism proceeds in three stages:

**GENERAL S<sub>E</sub>AR MECHANISM STEPS****1. Generation of Strong Electrophile (E<sup>+</sup>):**

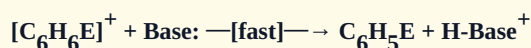
- **Nitration:**  $\text{HNO}_3 + 2\text{H}_2\text{SO}_4 \rightleftharpoons \text{NO}_2^+$  (Nitronium) +  $\text{H}_3\text{O}^+$  +  $2\text{HSO}_4^-$
- **Halogenation:**  $\text{Br}_2 + \text{FeBr}_3 \rightleftharpoons [\text{FeBr}_4]^- + \text{Br}^+$
- **Friedel-Crafts Alkylation/Acylation:**  $\text{R-Cl} + \text{AlCl}_3 \rightleftharpoons [\text{AlCl}_4]^- + \text{R}^+$  |  $\text{RCOCl} + \text{AlCl}_3 \rightleftharpoons [\text{AlCl}_4]^- + \text{R-C}\equiv\text{O}^+$  (Acylium)

**2. Attack on Aromatic Ring & Formation of Arenium Ion (σ-Complex / Wheland Intermediate):**

The electrophile attacks the π-electron cloud (slow, rate-determining step). The aromaticity is temporarily disrupted to produce a resonance-stabilized cyclo-hexadienyl carbocation delocalized over ortho and para positions:

**3. Deprotonation (Aromatization):**

A weak base in the medium ( $\text{HSO}_4^-$ ,  $[\text{AlCl}_4]^-$ ) rapidly removes the  $\text{sp}^3$  geminal proton, restoring the stable 6π aromatic sextet (fast step):

**PART 3 MECHANISM & APPLICATIONS OF MANDATED NAMED REACTIONS****1. Suzuki-Miyaura Cross-Coupling Reaction**

**Definition:** The palladium-catalyzed cross-coupling of an organoboron reagent (typically an aryl, alkenyl, or alkyl boronic acid,  $\text{R}'\text{-B(OH)}_2$ ) with an organic halide or pseudohalide ( $\text{R-X}$ , where  $\text{X} = \text{I, Br, Cl, OTf}$ ) in the presence of a base to form carbon-carbon (C-C) biaryl or substituted alkene bonds.

**GENERAL REACTION****CATALYTIC CYCLE MECHANISM**

1. **Oxidative Addition:** The active, coordinatively unsaturated palladium(0) catalyst ( $\text{Pd}^0\text{L}_2$ ) inserts into the carbon-halogen bond of  $\text{R-X}$  to produce a square-planar organopalladium(II) complex:  $\text{L}_2\text{Pd}^0 + \text{R-X} \rightarrow \text{R-Pd}^{\text{II}}\text{L}_2\text{-X}$  (Rate-limiting step for chlorides/bromides).
2. **Crucial Role of the Base:** Organoboron species are neutral and insufficiently nucleophilic to transfer their organic group directly. The base ( $\text{OH}^-$ ,  $\text{CO}_3^{2-}$ ,  $\text{OR}^-$ ) coordinates to the boron atom, forming a negatively charged, highly nucleophilic organoborate complex:  $\text{R}'\text{-B(OH)}_2 + \text{OH}^- \rightarrow [\text{R}'\text{-B(OH)}_3]^-$ . Alternatively, the base exchanges with the halide on palladium to form  $\text{R-Pd}^{\text{II}}\text{L}_2\text{-OH}$ .
3. **Transmetalation:** The activated organoborate transfers its  $\text{R}'$  group to the palladium(II) center, displacing the halide and releasing non-toxic borate byproducts:  $\text{R-Pd}^{\text{II}}\text{L}_2\text{-X} + [\text{R}'\text{-B(OH)}_3]^- \rightarrow \text{R-Pd}^{\text{II}}\text{L}_2\text{-R}' + \text{B(OH)}_3 + \text{X}^-$ .
4. **Reductive Elimination:** The cis-di-organopalladium complex undergoes intramolecular reductive elimination to forge the new  $\text{R-R}'$  single bond and regenerate the active  $\text{Pd}^0\text{L}_2$  catalyst for the next cycle:  $\text{R-Pd}^{\text{II}}\text{L}_2\text{-R}' \rightarrow \text{R-R}' + \text{Pd}^0\text{L}_2$ .

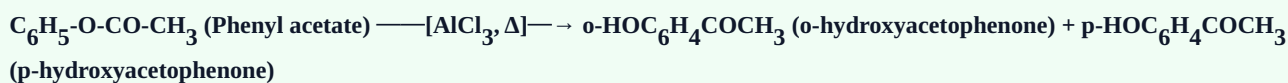
**Industrial & Synthetic Applications:** Synthesis of unsymmetrical biaryls in pharmaceutical manufacturing (e.g., active pharmaceutical ingredients like Losartan [antihypertensive], Boscalid [fungicide]); construction of conjugated organic polymers, OLED semiconductors, and liquid crystal materials. Tolerates wide functional groups ( $-\text{OH}$ ,  $-\text{NH}_2$ ,  $-\text{NO}_2$ ,  $-\text{CHO}$ ) and runs under mild, aqueous conditions with low toxicity.



## 2. Fries and Photo-Fries Rearrangement

### A. Classical Thermal Fries Rearrangement

**Definition:** The transformation of phenolic esters (aryl esters) into ortho- and para-hydroxy aryl ketones upon heating with a Lewis acid catalyst (typically anhydrous  $\text{AlCl}_3$ ).



#### THERMAL MECHANISM & TEMPERATURE SELECTIVITY

- 1. Complexation & Acylium Ion Formation:**  $\text{AlCl}_3$  coordinates to the ester phenolic oxygen, followed by heterolytic cleavage of the acyl-oxygen bond to release a free resonance-stabilized acylium cation ( $\text{CH}_3\text{-C}\equiv\text{O}^+ \leftrightarrow \text{CH}_3\text{-C}^+=\text{O}$ ) and an aluminum phenolate complex:  $\text{Ar-O-AlCl}_3^-$ .
- 2. Electrophilic Aromatic Attack:** The acylium ion acts as an electrophile, attacking the phenolate ring at either the ortho or para position via an electrophilic aromatic substitution pathway ( $\text{S}_{\text{E}}\text{Ar}$ ).
- 3. Thermodynamic vs. Kinetic Control:**
  - *Low Temperature* ( $\sim 60^\circ\text{C}$ ): Favors the **para-isomer** (less sterically hindered, kinetically controlled product).
  - *High Temperature* ( $> 160^\circ\text{C}$ ): Favors the **ortho-isomer** (thermodynamically controlled, stabilized by an exceptionally strong six-membered intramolecular hydrogen-bonded ring with the adjacent phenolic  $-\text{OH}$  group).

### B. Photo-Fries Rearrangement

**Definition & Distinctions:** The rearrangement of aryl esters to ortho- and para-hydroxy aryl ketones irradiated with ultraviolet light ( $h\nu = 254 \text{ nm}$ ) in the *complete absence of a Lewis acid catalyst*.

- **Mechanism (Intramolecular Radical-in-Cage):** Photoexcitation excites the aryl ester to an  $\text{S}_1$  or  $\text{T}_1$  state, prompting homolytic cleavage of the  $\text{ArO-COR}$  bond. This generates an aryloxy radical ( $\text{ArO}^\bullet$ ) and an acyl radical ( $\text{RCO}^\bullet$ ) trapped within a solvent cage.
- The radicals recombine within the solvent cage at the ortho or para positions of the phenoxy ring, followed by tautomerization to yield ortho- and para-acylphenols. Phenol ( $\text{ArOH}$ ) is often obtained as a decarbonylation/cage-escape byproduct.
- **Advantage:** Can be performed on substrates bearing acid-sensitive functional groups where Lewis acids would decompose the substrate.

## 3. Wagner-Meerwein Rearrangement

**Definition:** A 1,2-migration of an alkyl, aryl, or hydride group from an adjacent carbon to an electron-deficient carbocation center, generating a thermodynamically more stable carbocation.

#### DRIVING FORCE & MECHANISM

- 1. Generation of Carbocation:** Acid-catalyzed dehydration of alcohols or heterolytic loss of a leaving group from alkyl halides generates an initial carbocation:  $\text{R-CH(OH)-C(CH}_3)_3 + \text{H}^+ \rightarrow \text{R-CH}^+-\text{C(CH}_3)_3 + \text{H}_2\text{O}$ .
  - 2. 1,2-Shift:** The migrating group transfers with its bonding electron pair via a cyclic, non-classical 3-center-2-electron (3c-2e) bridged transition state:  

$$\text{R-CH}^+-\text{C(CH}_3)_3 \xrightarrow{[1,2\text{-methyl shift}]} \text{R-CH(CH}_3\text{)-C}^+(\text{CH}_3)_2$$
 (Conversion of a secondary carbocation into a more stable tertiary carbocation).
  - 3. Deprotonation:** Loss of a proton yields the most highly substituted, thermodynamically stable alkene (Saytzeff's rule).
- Primary Driving Forces:** (1) Relief of severe carbocyclic ring strain (e.g., cyclobutyl to cyclopentyl, or bicyclic terpene rearrangement); (2) Thermodynamic stabilization ( $1^\circ \rightarrow 2^\circ \rightarrow 3^\circ$  or resonance-stabilized allylic/benzylic cations).

**Classic Example in Terpene Chemistry:** Conversion of isoborneol / camphene hydrochloride to bornyl chloride (pinene-camphene rearrangements); biosynthesis of cholesterol and steroids via squalene oxide cascade cyclization.

#### 4. Umpolung Reactions (Corey-Seebach 1,3-Dithiane Synthesis)

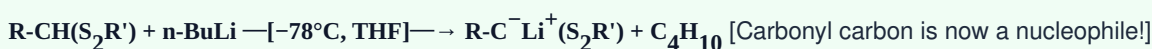
**Definition:** "Umpolung" (German for *polarity reversal*) is any chemical modification that reverses the normal, inherent alternating polarity and chemical reactivity of a functional group. Normally, the carbonyl carbon is electrophilic (electron-deficient,  $d^1$  donor acceptor pattern:  $R-C^{\delta+}=O$ ). Umpolung transforms it into a powerful nucleophile ( $d^1$  nucleophile / carbanion).

##### COREY-SEEBACH REACTION SEQUENCE

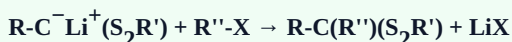
1. **Thioacetal Formation:** Aldehyde is reacted with propane-1,3-dithiol catalyzed by  $BF_3 \cdot Et_2O$  to produce a 1,3-dithiane:



2. **Deprotonation (Polarity Inversion):** The hydrogen at C-2 of 1,3-dithiane is weakly acidic ( $pK_a \approx 31$ ). Treatment with *n*-butyllithium (*n*-BuLi) in THF at  $-78^\circ C$  deprotonates C-2, producing 2-lithio-1,3-dithiane. The negative charge is stabilized by polarizable adjacent sulfur 3d-orbitals and inductive effects:



3. **Nucleophilic Alkylation:** The carbanion attacks an electrophile (alkyl halide  $R''-X$ , epoxide, or carbonyl compound):



4. **Hydrolytic Deprotection (Regeneration of Carbonyl):** Mild oxidative cleavage using  $HgCl_2 / CaCO_3$  or  $I_2 / NaHCO_3$  regenerates the carbonyl function, yielding a ketone:



**Other Examples of Umpolung:** The *Benzoin Condensation* (where cyanide ion  $CN^-$  acts as a catalyst, adding to benzaldehyde to convert the electrophilic carbonyl carbon into a nucleophilic carbanion enaminol intermediate that attacks a second benzaldehyde molecule).

#### 5. Favorskii Rearrangement

**Definition:** The skeletal rearrangement of  $\alpha$ -haloketones (bearing at least one  $\alpha'$ -hydrogen on the other side of the carbonyl) in the presence of base (hydroxide, alkoxides, amines) to yield carboxylic acid derivatives (acids, esters, or amides).

##### GENERAL REACTION



##### CYCLOPROPANONE INTERMEDIATE (LOFTFIELD) MECHANISM

1. **Enolate Formation:** Base abstracts an acidic  $\alpha'$ -proton from the non-halogenated side of the carbonyl to form an enolate ion:



2. **Intramolecular Displacement (Cyclization):** The carbanionic enolate carbon performs an intramolecular nucleophilic substitution ( $S_N2$ ) displacing the halide ion ( $X^-$ ), generating a strained **cyclopropanone intermediate**.

3. **Nucleophilic Addition & Ring Opening:** The alkoxide nucleophile ( $R''O^-$ ) attacks the electrophilic carbonyl carbon of the cyclopropanone, forming a tetrahedral oxyanion intermediate.

4. Relief of intense ring strain drives the collapse of the tetrahedral intermediate with cleavage of the cyclopropane ring at the bond that generates the **more stable carbanion** (more substituted/stabilized). Subsequent proton capture from the solvent yields the final rearranged ester.

**Quasi-Favorskii Rearrangement:** When the  $\alpha$ -haloketone lacks an enolizable  $\alpha'$ -hydrogen (e.g., in 1-bromobicyclo[2.2.2]octan-2-one), the reaction proceeds via direct nucleophilic addition of hydroxide to the carbonyl



followed by a **semi-benzilic 1,2-shift** with concerted halide ejection.

**Synthetic Utility:** Highly prized for achieving **ring contraction** in cyclic ketones (e.g., 2-chlorocyclohexanone cleanly rearranges with base to cyclopentanecarboxylic acid ester); used in the total synthesis of complex strained cage hydrocarbons like *Cubane*.

## PART 4 EFFLUENT ANALYSIS, POLLUTION CONTROL & OZONE DEPLETION

### 1. Chemical Analysis of Gaseous Effluents

- **Oxides of Sulphur ( $\text{SO}_x$  /  $\text{SO}_2$ ) — West-Gaeke Method:** Air is bubbled through 0.04 M potassium tetrachloromercurate ( $\text{K}_2[\text{HgCl}_4]$ ) to form dichlorosulphitomercurate complex:  $\text{SO}_2 + \text{H}_2\text{O} + [\text{HgCl}_4]^{2-} \rightarrow [\text{HgCl}_2(\text{SO}_3)]^{2-} + 2\text{H}^+ + 2\text{Cl}^-$ . Reacted with formaldehyde and bleached pararosaniline hydrochloride to form a brilliant magenta-red pararosaniline methylsulfonic acid dye, measured at  $\lambda = 548 \text{ nm}$ .
- **Oxides of Nitrogen ( $\text{NO}_x$ ) — Jacobs-Hochheiser / Saltzman Method:** Nitric oxide ( $\text{NO}$ ) is oxidized to  $\text{NO}_2$  using  $\text{CrO}_3$  /  $\text{KMnO}_4$ .  $\text{NO}_2$  is absorbed in a solution of sulfanilic acid and N-(1-naphthyl)-ethylenediamine dihydrochloride (NEDA) under acidic conditions, forming an azo dye measured spectrophotometrically at  $\lambda = 540 \text{ nm}$ . Continuous monitoring employs gas-phase *chemiluminescence* ( $\text{NO} + \text{O}_3 \rightarrow \text{NO}_2^* + \text{O}_2 \rightarrow \text{NO}_2 + \text{hv}$ ).
- **Hydrogen Sulphide ( $\text{H}_2\text{S}$ ) — Methylene Blue Method:**  $\text{H}_2\text{S}$  gas is fixed by bubbling through alkaline zinc acetate or cadmium hydroxide suspension, forming  $\text{ZnS}$  /  $\text{CdS}$  precipitate. Reacted with N,N-dimethyl-p-phenylenediamine sulfate in acidic  $\text{FeCl}_3$  to yield methylene blue dye, measured colorimetrically at  $\lambda = 670 \text{ nm}$ .

### 2. Chemical Analysis of Effluent Liquid Streams: DO, BOD, and COD

- **Dissolved Oxygen (DO) by Winkler's Method:** Manganese sulfate and alkaline-iodide-azide ( $\text{NaOH} + \text{KI} + \text{NaN}_3$ ) react with DO to yield brown basic manganese oxide:  $2\text{Mn}(\text{OH})_2 + \text{O}_2 \rightarrow 2\text{MnO}(\text{OH})_2$ . Acidified with conc.  $\text{H}_2\text{SO}_4$  to liberate iodine:  $\text{MnO}(\text{OH})_2 + 2\text{I}^- + 4\text{H}^+ \rightarrow \text{Mn}^{2+} + \text{I}_2 + 3\text{H}_2\text{O}$ . The liberated  $\text{I}_2$  is titrated against standard  $\text{Na}_2\text{S}_2\text{O}_3$  with starch indicator:

$$\text{DO (mg/L)} = (\text{V}_{\text{thiosulfate}} \times \text{N} \times 8 \times 1000) / \text{V}_{\text{sample}}$$

- **Biochemical Oxygen Demand (BOD):** Measure of dissolved oxygen required by aerobic microbes to biologically decompose biodegradable organic matter over **5 days at 20°C ( $\text{BOD}_5$ )**:

$$\text{BOD}_5 \text{ (mg/L)} = [(\text{DO}_0 - \text{DO}_5) - (\text{B}_0 - \text{B}_5)(1 - 1/P)] \times P$$

- **Chemical Oxygen Demand (COD):** Measure of total oxygen needed to chemically oxidize all organic matter (biodegradable + non-biodegradable) using standard potassium dichromate (**0.25 N  $\text{K}_2\text{Cr}_2\text{O}_7$** ) in concentrated  $\text{H}_2\text{SO}_4$  with  $\text{Ag}_2\text{SO}_4$  (catalyst) and  $\text{HgSO}_4$  (masks chloride interference) under 2-hour reflux. Back-titrated with standard ferrous ammonium sulfate (FAS) using ferroin indicator (blue-green to reddish-brown):

$$\text{COD (mg/L)} = [(\text{V}_{\text{blank}} - \text{V}_{\text{sample}}) \times \text{N}_{\text{FAS}} \times 8 \times 1000] / \text{V}_{\text{sample}}$$

*Relationship: COD is always greater than BOD ( $\text{COD} > \text{BOD}$ ). Normal domestic sewage has  $\text{BOD}/\text{COD} \approx 0.5 - 0.7$ .*

### 3. Comprehensive Control of Environmental Pollution

- **Air Pollution Control:**
  - *Particulate Matter:* Electrostatic Precipitators (ESP; >99% efficiency for submicron particles via corona discharge), Cyclone Separators (centrifugal force for particles >10  $\mu\text{m}$ ), Baghouses (fabric filtration), and Venturi Scrubbers.

- *Gaseous Pollutants*: Flue Gas Desulfurization (FGD with wet limestone:  $\text{CaCO}_3 + \text{SO}_2 + \frac{1}{2}\text{O}_2 + 2\text{H}_2\text{O} \rightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$  [gypsum]); Selective Catalytic Reduction (SCR for  $\text{NO}_x$ :  $4\text{NO} + 4\text{NH}_3 + \text{O}_2 \xrightarrow{[\text{V}_2\text{O}_5\text{-TiO}_2]} 4\text{N}_2 + 6\text{H}_2\text{O}$ ); Three-Way Catalytic Converters (Pt-Pd-Rh on cordierite oxidizing  $\text{CO/HC}$  to  $\text{CO}_2/\text{H}_2\text{O}$  and reducing  $\text{NO}_x$  to  $\text{N}_2$ ).

#### • Wastewater Pollution Control:

- *Primary (Physical)*: Screening, grit removal chambers, and primary sedimentation (gravity settling of suspended solids).
- *Secondary (Biological)*: Activated Sludge Process (aerobic microbial flocs oxidize organics, followed by secondary clarifiers and recycled activated sludge [RAS]); Trickling Filters.
- *Tertiary (Physicochemical)*: Coagulation-flocculation using Alum, granular activated carbon adsorption, Reverse Osmosis (RO) desalinization, and pathogen disinfection via Chlorination or UV radiation.

## 4. Depletion of the Stratospheric Ozone Layer

### PHOTOCHEMICAL MECHANISM & CATALYTIC DEPLETION

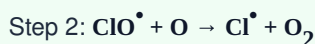
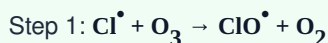
#### Chapman Natural Steady-State Cycle:

1.  $\text{O}_2 + h\nu (\lambda < 242 \text{ nm}) \rightarrow \text{O} + \text{O}$
2.  $\text{O} + \text{O}_2 + \text{M} \rightarrow \text{O}_3 + \text{M}$
3.  $\text{O}_3 + h\nu (\lambda \approx 200\text{--}310 \text{ nm}) \rightarrow \text{O}_2 + \text{O}$
4.  $\text{O}_3 + \text{O} \rightarrow 2\text{O}_2$

#### Anthropogenic Catalytic Depletion by CFCs (Freons):

UV photolysis releases atomic chlorine:  $\text{CF}_2\text{Cl}_2 + h\nu (\lambda < 220 \text{ nm}) \rightarrow \text{CF}_2\text{Cl}^\bullet + \text{Cl}^\bullet$ .

Catalytic chain cycle:



Net:  $\text{O}_3 + \text{O} \rightarrow 2\text{O}_2$ . One chlorine radical destroys over **100,000 ozone molecules** before termination.

**Antarctic Ozone Hole & Polar Stratospheric Clouds (PSCs)**: Winter polar vortex drops temperatures below  $-80^\circ\text{C}$ , producing ice/nitric acid clouds (PSCs) that catalyze heterogeneous conversion of inert reservoirs (**HCl, ClONO<sub>2</sub>**) into photolabile molecular chlorine:  $\text{HCl(s)} + \text{ClONO}_2(\text{g}) \rightarrow \text{Cl}_2(\text{g}) + \text{HNO}_3(\text{s})$ . Spring sunlight triggers massive photolytic release of  $\text{Cl}^\bullet$  radicals.

**Environmental Consequences & Regulatory Treaties**: Increased UV-B leads to cataracts, non-melanoma/melanoma skin cancer (via pyrimidine dimer mutation), immunosuppression, and massive decline in marine phytoplankton. Mitigated internationally by the legally binding **Montreal Protocol (1987)** phasing out ODS, and the **Kigali Amendment (2016)** phasing down hydrofluorocarbons (HFCs).